

## When does the Stock Market bottom in a Recession?

There have been many calls by economists for a recession in the first quarter of 2023. Does the stock market bottom before a recession? Or during? Or after? Lets take a look.

First, lets see where we are at now. The Gross Domestic Product in the United States hit its high in the Fourth Quarter of 2021.<sup>1</sup> It doesn't mean we are in a recession now (the official dating of recession is the National Bureau of Economic Research (a non-partisan think tank group of business and labor leaders, as well as university-based economists who were committed to uncovering and disseminating important facts about the economy of the United States)<sup>2</sup>. So, as we know it right now, the peak of the economy was in the last quarter of 2021.

So when does the economy hit its bottom? Below is a chart of each recession since the end of World War 2. The average time when the economy hits its bottom or trough is 10.41 months.

### ~~S3 File. Results of balance for alternatives in designed questionnaire.~~

| <del>GDP Block</del>    | <del>Scenario</del>    | <del>GDP Trough</del> | <del>GDPA Trough</del>   | <del>Peak to Trough Food-drug</del> | <del>Monitoring</del> | <del>Bleeding</del> | <del>Stroke</del> | <del>AMI</del> | <del>Cost</del> | <del>Dominant checking*</del> |
|-------------------------|------------------------|-----------------------|--------------------------|-------------------------------------|-----------------------|---------------------|-------------------|----------------|-----------------|-------------------------------|
| <del>Peak Month 1</del> | <del>Peak Year 1</del> | <del>Month A</del>    | <del>Trough Year 1</del> | <del>in month 1</del>               | <del>3</del>          | <del>3</del>        | <del>2</del>      | <del>0</del>   | <del>0</del>    | <del>3</del>                  |
|                         |                        | B                     | 1                        | 0                                   | 0                     | 0                   | 3                 | 1              | 3               | 3                             |
|                         | 2                      | A                     | 0                        | 1                                   | 2                     | 0                   | 0                 | 0              | 1               | 1                             |
|                         |                        | B                     | 0                        | 1                                   | 1                     | 1                   | 0                 | 0              | 2               | 2                             |
|                         | 3                      | A                     | 0                        | 0                                   | 1                     | 1                   | 1                 | 2              | 2               | 1                             |
| Febru                   | 2020                   | April                 | 2020                     | 0                                   |                       | 1                   | 2                 | 2              | 1               | 3                             |
| ary                     | 2020                   | B                     | 1                        |                                     | 2                     |                     |                   |                |                 |                               |
| Dece                    | 2007                   |                       | 2009                     |                                     | 3                     | 2                   | 3                 | 3              | 3               | 3                             |
| mber                    | 4                      | June                  | 0                        | 170                                 |                       |                     |                   |                |                 |                               |
|                         |                        | B                     | 0                        | 0                                   | 1                     | 3                   | 3                 | 2              | 0               | 1                             |
|                         | 5                      | A                     | 1                        | 0                                   | 1                     | 3                   | 2                 | 2              | 3               | 3                             |
|                         |                        | B                     | 1                        | 0                                   | 0                     | 2                   | 2                 | 3              | 0               | 1                             |
|                         | 6                      | A                     | 1                        | 1                                   | 0                     | 0                   | 3                 | 3              | 1               | 3                             |
|                         |                        | B                     | 1                        | 0                                   | 3                     | 3                   | 1                 | 1              | 2               | 3                             |
| Marc                    | 2001                   | Novem                 | 2001                     |                                     | 2                     | 0                   | 1                 | 0              | 2               | 2                             |
| h                       | 7                      | ber                   | 0                        | 71                                  |                       |                     |                   |                |                 |                               |
|                         |                        | B                     | 0                        | 1                                   | 3                     | 0                   | 0                 | 0              | 1               | 1                             |
|                         | 1990                   | March                 | 1991                     |                                     | 0                     | 3                   | 0                 | 1              | 0               | 2                             |
| July                    | 8                      | A                     | 1                        | 80                                  |                       |                     |                   |                |                 |                               |
|                         |                        | Novem                 | 1982                     |                                     | 1                     | 2                   | 1                 | 3              | 3               | 5                             |
| July                    | 1981                   | ber                   | 0                        | 171                                 |                       |                     |                   |                |                 |                               |

<sup>1</sup> <https://www.bea.gov/data/gdp/gross-domestic-product>

<sup>2</sup> <https://www.nber.org/about-nber/history>

Source: <https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions>

Now, lets review when the stock market bottoms. Does the stock market bottom after the economy starts to rebound? No, markets (including the stock market) are forward looking - they take what they know about the future and discount it to today.

So back to stock markets, when does the stock market on average bottom in a recession? On average, since the end of World War 2, the stock market bottoms 4.5 months before the economy does.

<sup>3</sup> <https://www.nber.org/news/business-cycle-dating-committee-announcement-july-19-2021>

| <u>Peak Month</u> | <u>Peak Year</u> | <u>Trough Month</u> | <u>Trough Year</u> | <u>Peak Day</u>  | <u>Trough Day</u> | <u>Before the Economy</u> |
|-------------------|------------------|---------------------|--------------------|------------------|-------------------|---------------------------|
| <u>February</u>   | <u>2020</u>      | <u>April</u>        | <u>2020</u>        | <u>24-Feb-20</u> | <u>23-Mar-20</u>  | <u>1</u>                  |
| <u>December</u>   | <u>2007</u>      | <u>June</u>         | <u>2009</u>        | <u>12-Oct-07</u> | <u>09-Mar-09</u>  | <u>3</u>                  |
| <u>March</u>      | <u>2001</u>      | <u>November</u>     | <u>2001</u>        | <u>01-Sep-00</u> | <u>21-Sep-21</u>  | <u>2</u>                  |
| <u>July</u>       | <u>1990</u>      | <u>March</u>        | <u>1991</u>        | <u>13-Jul-90</u> | <u>12-Oct-90</u>  | <u>6</u>                  |
| <u>July</u>       | <u>1981</u>      | <u>November</u>     | <u>1982</u>        | <u>12-Jun-81</u> | <u>06-Aug-82</u>  | <u>5</u>                  |
| <u>January</u>    | <u>1980</u>      | <u>July</u>         | <u>1980</u>        | <u>08-Feb-80</u> | <u>18-Apr-80</u>  | <u>3</u>                  |
| <u>November</u>   | <u>1973</u>      | <u>March</u>        | <u>1975</u>        | <u>12-Oct-73</u> | <u>04-Oct-74</u>  | <u>6</u>                  |
| <u>December</u>   | <u>1969</u>      | <u>November</u>     | <u>1970</u>        | <u>01-Jan-70</u> | <u>22-May-70</u>  | <u>5</u>                  |
| <u>April</u>      | <u>1960</u>      | <u>February</u>     | <u>1961</u>        | <u>10-Jun-60</u> | <u>21-Oct-60</u>  | <u>5</u>                  |
| <u>August</u>     | <u>1957</u>      | <u>April</u>        | <u>1958</u>        | <u>12-Jul-57</u> | <u>20-Dec-57</u>  | <u>5</u>                  |
| <u>July</u>       | <u>1953</u>      | <u>May</u>          | <u>1954</u>        | <u>07-Aug-53</u> | <u>11-Sep-53</u>  | <u>9</u>                  |
| <u>November</u>   | <u>1948</u>      | <u>October</u>      | <u>1949</u>        | <u>22-Oct-48</u> | <u>10-Jun-49</u>  | <u>4</u>                  |
|                   |                  |                     |                    |                  | <u>average</u>    | <u>4.5</u>                |
|                   |                  |                     |                    |                  | <u>median</u>     | <u>5</u>                  |

So if we have had a couple of quarters of negative growth in the economy already in 2022 and if we have a recession called for in 2023 and if the market bottoms about 4 and a half months before the economy hits its trough, we are somewhere in the ballpark at a market bottom.

Now, this is a bunch of math and data and history and each market cycle is different, but I wanted to present to you that to me, for a variety of reasons (not just the ones I have presented here, the charts also point to a short term bottoming process), we are probably close to the bottom of the market in this market cycle.

Let me know if you have any questions or concerns. Please feel free to forward this to anyone you think may find it interesting.

Thank you

Don Food-drug indicates food-drug interaction; monitoring indicates the frequency of blood monitoring; bleeding indicates the risk of major bleeding; stroke indicates the risk of stroke (ischemic/hemorrhagic) or systemic embolism; AMI indicates the risk of acute myocardial infarction; cost indicates monthly out-of-pocket cost. RMB indicates Ren Min Bi, the unit of Chinese currency.

\*The value in dominant checking represents the number of dominated attributes for one alternative in specific scenarios. All levels of the attributes would be the same if the values were all 0 in the same scenario (which is impossible for trade-offs). One alternative would be dominated if its value was greater than 0, and the other one was 0.

Levels for antidote and food-drug interaction is indicated by “Yes” if 1, “No” if 0; levels for the frequency of blood monitoring is indicated by “no need” if 0, “every 6 months” if 1, “every 3 months” if 2, “every 1 month” if 3; levels for the risk of major bleeding is indicated by “0.7%” if 0, “3.1%” if 1, “5.5%” if 2, “7.8%” if 3; levels for the risk of stroke or systemic embolism is indicated by “0.6%” if 0, “3.2%” if 1, “5.8%” if 2, “8.4%” if 3; levels for the risk of acute myocardial infarction is indicated by “0.2%” if 0, “1.0%” if 1, “1.8%” if 2, “2.5%” if 3; levels for out-of-pocket is indicated by “0 RMB” if 0, “120 RMB” if 1, “240 RMB” if 2, “360 RMB” if 3.